# ELENA MATYUSHENKO

Senior Generative AI Engineer

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**Civitai: 121 published models | 111K+ downloads | 13K+ likes | 1,300+ followers**

## PROFESSIONAL SUMMARY

Senior Generative AI engineer building AI-powered products end-to-end — from training foundation diffusion models from scratch to optimizing inference pipelines and shipping full-stack web platforms with real users. Led a distributed team of 8 developers (through Jan 2026) and now run production solo with AI-augmented engineering. Trained Samay — a text-to-image foundation model built from zero on a single desktop GPU (288.65M flow-matching diffusion transformer + own 46.0M 16-channel VAE, 1.12M self-curated images, no borrowed pretrained weights). Built and shipped NexTry (AI virtual try-on platform, 3,100+ users, 16+ generator tools), secondhand.fi (AI resale marketplace, live), BotBazar (AI bot marketplace, in beta), and TaskPilot (open-source AI agent framework). Published 121 models on Civitai with 111K+ total downloads and 1,300+ followers.

## CORE SKILLS

### Generative AI — Model Training & Fine-tuning

**Foundation model training (Samay, 2nd generation):** text-to-image model trained from zero — 288.65M-parameter flow-matching diffusion transformer with dual-stream text-image conditioning, 2D axial RoPE, register tokens, EMA weights and REPA representation alignment; 1.12M self-curated images, zero borrowed pretrained weights, frozen Qwen text encoder; trained on one ~32 GB desktop GPU — 512px stage complete, 768px stage training now, 1024px+ next

**VAE autoencoders:** trained a custom 16-channel latent autoencoder with self-attention (46.0M params, ~12:1 compression) to replace the 8-channel quality ceiling — GSC skip-connections, sharpness-based data filtering, controlled GAN/adversarial component; latent-statistics and reconstruction-quality analysis

**Foundation model training (1st generation):** text-to-image diffusion transformer (DiT, ~202M params) trained from scratch to 470K+ steps at 512px — custom VAE latent space + Qwen2-7B text encoder + DiT denoiser; single-stream DiT architecture; edit/try-on variant with packshot reference tokens

**Dataset engineering:** industrial web scraping across ~30 sources (anti-bot / TLS-fingerprint bypass via curl_cffi, nodriver, yt-dlp); automated VTON triplet mining (SigLIP + YOLO11-pose + ArcFace); deduplication, quality filtering, and unified VLM-as-judge re-captioning (Qwen2.5-VL); streaming / mmap pipelines for datasets larger than RAM

**Training infrastructure:** custom pipelines in Python/PyTorch/Diffusers — dataloading, preprocessing, training loops, validation, hyperparameter search, logging; memory-tuned to fit a single card's VRAM ceiling (SDPA kernels, gradient checkpointing, batch/accumulation sweeps)

**Diffusion stacks:** SD 1.5/2.1/3, SDXL, FLUX, FLEX, HiDream, Qwen

**Fine-tuning methods:** LoRA / LyCORIS (LoHa, LoKr, LoCon), DreamBooth, Textual Inversion

**Training specializations:** style transfer, identity/character consistency, product realism, photorealism

### Computer Vision — Control & Pipelines

**ComfyUI:** production-ready workflows, sampler/scheduler cross-testing, 2–5x inference speedups

**Control adapters:** ControlNet, IP-Adapter, PuLID

**Segmentation/detection:** SAM / SAM3 (Segment Anything), YOLO

**Vision-language:** Qwen-VL / Qwen2.5-VL as captioner and as VLM-as-judge in data pipelines

**Image processing:** inpainting/outpainting, img2img, background replacement, product placement

**Video:** Luma, Kling, AnimateDiff, Runway, Hunyuan Video, WAN

### Full-Stack Development

**Frontend:** Next.js 14/15 (App Router), React 19, TypeScript, Tailwind CSS, SCSS, Radix UI / shadcn, GSAP

**Backend:** NestJS, Node.js, Express, REST API design, Swagger/OpenAPI, WebSocket real-time

**Database:** PostgreSQL, Prisma ORM, SQLite, Redis

**Auth:** Google OAuth 2.0, JWT, next-auth, email/password, API key auth, 2FA (TOTP)

**Payments:** Stripe (Checkout, Connect, Elements), escrow system, commission logic

### AI Agent Development

**Agent frameworks:** custom TypeScript agent loop, tool registry, skill-based access control

**LLM integration:** 16+ providers (OpenAI, Anthropic, Groq, Ollama), function calling, prompt engineering

**Security:** command sandboxing, dangerous command blocking, approval workflows, encrypted settings

### Leadership & Team Management

**Team coordination:** led a distributed team of 8 developers (through Jan 2026) — task allocation, code review, multi-developer Git workflows; now solo with AI-augmented engineering

**AI-augmented engineering:** leveraging AI coding tools (Claude Code, Cursor) to accelerate development velocity

**Product management:** requirements gathering, feature prioritization, sprint planning, technical decision-making

**Process design:** code style guides, commit conventions, documentation standards, CI/CD pipelines

### Infrastructure & Deployment

**Containers:** Docker, docker-compose, multi-service orchestration

**Cloud:** AWS, Google Cloud, Azure, Lambda Labs, Scaleway, Nebius

**Web servers:** Caddy (auto-TLS), Nginx

**CI/CD:** Git workflows, custom linters, architecture validation

**Optimization:** GPU inference tuning, batching, throughput/latency optimization

### Data & Tooling

**Core:** Python, PyTorch, Hugging Face Diffusers, Jupyter, NumPy, pandas, scikit-learn, SQL, Linux/Bash

## PROFESSIONAL EXPERIENCE

### Founder & Lead AI Engineer — NexTry / secondhand.fi / BotBazar / TaskPilot | 2024-Present

*Founder building AI products end-to-end — foundation model training, virtual try-on platform, AI resale marketplace, and open-source agent framework*

- Samay — text-to-image foundation model trained from scratch: own 16-channel VAE (46.0M params, ~12:1 compression) + 288.65M flow-matching diffusion transformer, 1.12M self-curated images, zero borrowed pretrained weights, one ~32 GB desktop GPU; 512px stage done, 768px training now, 1024px+ next — VAE, latent statistics, transformer, sampler and data pipeline all authored in-house
- Earlier from-scratch models: a custom virtual try-on diffusion model (59K-pair dataset — reference/garment/target with masks, OpenPose, SAM) and a text-to-image diffusion transformer (DiT, ~202M, 470K+ steps) with a custom VAE and Qwen2 text encoder
- NexTry — AI virtual try-on platform (3,100+ users, 16+ generator tools): built and launched a Next.js 15 + React 19 product — virtual try-on, mannequin generation, virtual photoshoots, colorization, pose transfer, object placement; own try-on weights run in production behind a self-hosted ComfyUI + SAM3 pipeline, no third-party try-on API
- secondhand.fi — AI-powered resale marketplace for Finland (live, built solo in 3 months): 18 backend modules / 148 TypeScript source files on NestJS + Next.js 14 — AI listing auto-fill from one photo in ~3s (self-hosted Qwen-VL + GPT-4o-mini), Dutch auctions, Stripe escrow + Connect auto-payouts, Posti SmartPost shipping labels, WebSocket chat, disputes, business batch upload, and virtual try-on on real listings powered by our own model
- Designed and optimized ComfyUI production workflows — systematic sampler/scheduler cross-testing for 2–5x speedups (mannequin 7.5s → 2.9s, colorize 14.4s → 4.3s, restore 33.3s → 4.0s)
- Led a distributed team of 8 developers through Jan 2026 (code review, release cycles, multi-developer Git workflows); now running production solo with AI-augmented engineering (Claude Code, Cursor)
- BotBazar — AI bot marketplace (beta): architected a full-stack monorepo from scratch (Next.js 14 + NestJS + PostgreSQL + Redis) with Stripe escrow (10% commission) + Stripe Connect payouts, multi-role auth (Google OAuth, 2FA/TOTP, API keys), and an LLM content-moderation pipeline
- TaskPilot — open-source (MIT) security-first AI agent framework: TypeScript agent loop with tool registry, memory, command-security layer, skill-based access control (4 tiers), 16+ LLM providers, and a Docker execution sandbox

- CardAI — AI product-card generator for marketplaces: 4 marketplace-ready cards from a single product photo in ~60s; separate NestJS/Python backend + Next.js frontend, PostgreSQL + Redis, ComfyUI generation, OpenAI + Gemini
- Delivered additional platforms: Alumni Network — a private alumni community (NestJS + Prisma + PostgreSQL, JWT auth, email/moderation/audit, Next.js frontend, Docker + Caddy TLS) — and a headless-browser flight/price aggregator scraper (Playwright, JS-render bypass, no API keys)
- Deployed all products with Docker on GPU cloud (AWS/GCP/Azure/Lambda/Scaleway/Nebius); built a custom architecture linter enforcing layer rules, file-size limits, and import conventions

**Senior Generative AI Engineer — Freelance (Remote)** | 2021–2024

*Commercial AI model training and production pipeline development*

- Published 121 models on Civitai (111K+ downloads, 13K+ likes, 1,300+ followers) — LoRA/LyCORIS for fashion, product visuals, portraits, style systems
- Built end-to-end ComfyUI production pipelines: virtual try-on, mannequin generation, photoshoots, colorization, pose transfer, object placement
- Designed and prepared large-scale datasets: annotation/captioning guidelines, cleaning, deduplication, balancing, metadata schema
- Implemented training pipelines in Python/PyTorch/Diffusers: dataloading, preprocessing, training loops, validation, logging, config iteration
- Deployed models and inference APIs with Docker on GPU cloud (AWS/GCP/Azure/Lambda/Scaleway/Nebius), optimizing cost/performance for production

## EDUCATION

**PhD in Law** — analytical foundation applied to technical architecture, IP strategy, and compliance-aware product design

## LANGUAGES

Russian (Native) | English (Fluent)